Design and Development of a Bristle-Based Brush Bot for Comfort Room Cleaning

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INTRODUCTION

Background of the Study

Maintaining cleanliness in comfort rooms is important to ensure proper hygiene, safety, and a healthy environment. Poor sanitation can increase the risk of infectious diseases, especially in shared facilities. (World Health Organization, 2024). Bathroom surfaces can become sources of harmful microorganisms because fecal materials contain large numbers of pathogens that may spread on toilet surfaces during use (CDC, 2013). One study showed that as many as 10^{14} enteric pathogens can be excreted in a single bowel movement of 100 grams, while an individual may produce 100 to 200 grams of feces per day(Cortnery, et al. 2013) Bathrooms are also considered hazardous areas because contamination and accidents commonly occur there, especially in public and institutional restrooms (Kaivac, 2014; Tierno, 2015). Because of this, regular and effective cleaning of comfort rooms is necessary to maintain sanitation and prevent health risks (WHO, 2018).

Traditional cleaning methods such as using mops and brushes require physical effort, time, and continuous manual labor. These methods may not always provide consistent cleaning results, especially in hard-to-reach areas and tiled surfaces with stubborn dirt. Manual cleaning can also become tiring and inefficient when comfort rooms require frequent maintenance (Kaivac, 2014). Because of these challenges, there is a growing need for efficient and convenient cleaning technologies (Dragićević et al., 2016).

In recent years, robotics and automation have become useful in performing household cleaning tasks. Automated cleaning devices such as robot vacuums and floor-cleaning robots help reduce human effort while improving cleaning efficiency (Dragićević et al., 2016). The use of microcontrollers has become popular in robotics because of their affordability, flexibility, and accessibility for students and researchers. The use of microcontrollers has become popular in robotics because of their affordability, flexibility, and accessibility for students and researchers. These platforms allow developers to create simple and practical robotic systems for real-world applications (Draeger-IT, 2023).

This study focuses on the development of a High-Fidelity Prototype of an Arduino Uno-Based Brushbot Cleaning Robot. The prototype uses a large cleaning brush, a DC motor with uneven weight, and an Arduino-controlled motor system to create vibration and movement. The rotating and vibrating bristles allow the Brushbot to move across tiled surfaces while scrubbing dirt and debris.

Research Objectives

The purpose of this study was to design and develop a High-Fidelity Prototype of an Arduino Uno-Based Brushbot Cleaning Robot. Specifically, the study aims to:

1. How can a bristle-based brush bot be developed for comfort room cleaning?
2. How effective is the bristle-based brush bot on terms of mobility, dirt removal, and overall cleaning performance?
3. How can the bristle-based brush bot reduce the time, effort, and physical labor involved in manual comfort room cleaning?

Significance of the Study

This study is important because it introduces a simple and affordable robotic cleaning technology that may improve sanitation and reduce the difficulty of manual comfort room cleaning (World Health Organization, 2018).

Households
The Brushbot may provide households with a more convenient and efficient method of cleaning comfort rooms while reducing physical effort and cleaning time (JonrTech, 2025).

Schools and Communities
The study may promote awareness about proper sanitation and encourage the use of innovative technologies in solving everyday cleaning problems by aligning with global sanitation-and-health guidelines (Susana, 2025).

Future Researchers
This study may serve as a reference for future researchers who plan to develop automated cleaning systems and robotic technologies using low-cost platforms (Lee, 2021)

Sustainable Development Goals (SDGs)
This study supports Sustainable Development Goal 3 (Good Health and Well-Being) by promoting proper sanitation and hygiene practices and Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by encouraging technological innovation.

Scope and Delimitation
This study focuses on the design, development, and testing of a High-Fidelity Prototype of an Arduino Uno-Based Brushbot Cleaning Robot intended for comfort room floor cleaning. The prototype uses an Arduino Uno, DC motor, motor driver module, uneven motor weight, and rotating bristles to create vibration and movement for cleaning tiled surfaces.

The study is limited only to cleaning tiled comfort room floors and small dirt particles such as dust and debris. It does not include wall cleaning, toilet bowl cleaning, liquid waste removal, or advanced automation features such as obstacle detection, artificial intelligence, or wireless control systems. Testing of the prototype will only be conducted within a controlled environment to evaluate movement, stability, vibration strength, and cleaning performance. Commercialization, industrial deployment, and large-scale manufacturing are beyond the scope of this study.

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APENDIX A

Prototype

Utilize Tinkercad or any equivalent simulation/design platform to create the schematic and layout of your Arduino-based prototype.
Ensure that all components are properly labeled (e.g., Arduino board, sensors, actuators, power supply, wiring connections) for clarity and accuracy in documentation.

APPENDIX B

Coding

Insert your program codes in this section.